

ECE 667

Synthesis and Verification of Digital Circuits

Scheduling Algorithms
Analytical approach - ILP

Scheduling – a Combinatorial Optimization Problem

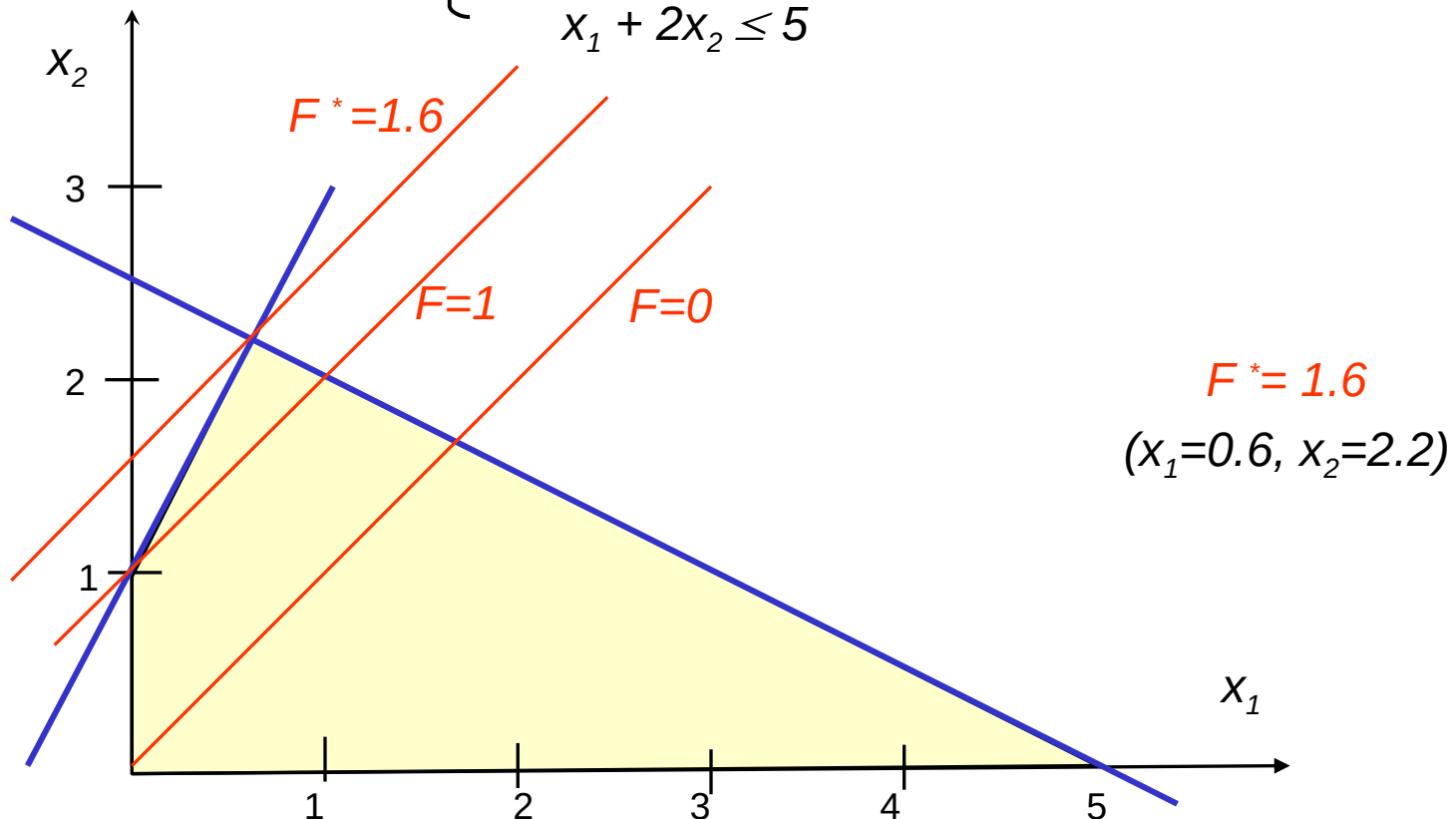
- NP-complete Problem
- Optimal solutions for special cases and for ILP
 - Integer linear program (ILP)
 - Branch and bound
- Heuristics
 - iterative Improvements, constructive
- Various versions of the problem
 - Minimum latency, unconstrained (ASAP)
 - Latency-constrained scheduling (ALAP)
 - Minimum latency under resource constraints (ML-RC)
 - Minimum resource schedule under latency constraint (MR-LC)
- If all resources are identical, problem is reduced to *multiprocessor scheduling* (Hu's algorithm)
 - In general, minimum latency multiprocessor problem is intractable under resource constraint
 - Under certain constraints (G(VE) is a tree), greedy algorithm gives optimum solution

Integer Linear Programming (ILP)

- Given:
 - integer-valued matrix $A_{m \times n}$
 - variables: $\mathbf{x} = (x_1, x_2, \dots, x_n)^T$
 - constants: $\mathbf{b} = (b_1, b_2, \dots, b_m)^T$ and $\mathbf{c} = (c_1, c_2, \dots, c_n)^T$
 - Minimize: $\mathbf{c}^T \mathbf{x}$
- subject to:
- $$\left\{ \begin{array}{l} \mathbf{A} \mathbf{x} \leq \mathbf{b} \\ \mathbf{x} = (x_1, x_2, \dots, x_n) \text{ is an integer-valued vector} \end{array} \right.$$
- If all variables are *continuous*, the problem is called linear (LP)
 - Problem is called *Integer* LP (ILP) if some variables x are integer
 - special case: 0,1 (binary) ILP

Linear Programming – example

- Variables: $x = [x_1, x_2]^T$
- Objective function: $\max F = -x_1 + x_2 = [-1 \ 1] [x_1, x_2]^T$
- Constraints:
$$\begin{cases} -2x_1 + x_2 \leq 1 \\ x_1 + 2x_2 \leq 5 \end{cases}$$



ILP Model of Scheduling

- Binary decision variables x_{il}

$x_{il} = 1$ if operation v_i starts in step l ,
otherwise $x_{il} = 0$

$i = 0, 1, \dots, n$ (operations)

$l = 1, 2, \dots, \lambda + 1$ (steps, with limit λ)

- Start time of each operation v_i is unique:

$$\sum_l x_{il} = 1, \quad i = 0, 1, \dots, n$$

Note:
$$\sum_l x_{il} = \sum_{l=t_i^S}^{l=t_i^L} x_{il}$$

where:

t_i^S = time of operation i computed with *ASAP*

t_i^L = time of operation i computed with *ALAP*

ILP Model of Scheduling - constraints

- Start time for v_i :

$$t_i = \sum_l l \cdot x_{il}$$

- Precedence relationships must be satisfied

$$\sum_l l \cdot x_{il} \geq \sum_l l \cdot x_{jl} + d_j, \quad i, j = 0, 1, \dots, n \quad : (v_j, v_i) \in E$$

- Resource constraints must be met

– let upper bound on number of resources of type k be a_k

$$\sum_{i:T(v_i)=k} \sum_{m=l-d_i+1}^l x_{im} \leq a_k, \quad k = 1, 2, \dots, n_{res}, \quad l = 1, 2, \dots, \bar{\lambda} + 1$$

Latency Minimization - Objective Function

- Function to be minimized: $F = c^T t$, where $t_i = \sum_l l \cdot x_{il}$
- Minimum latency schedule: $c = [0, 0, \dots, 1]^T$
 - $F = t_n = \sum_l l x_{nl}$
where t_n refers to the start time of *last node* (NOP) !
 - if sink has no mobility ($x_{n,s} = 1$), any feasible schedule is optimum
- ASAP: $c = [1, 1, \dots, 1]^T$
 - finds earliest start times for all operations $\sum_i \sum_l x_{il}$
 - or equivalently:

$$x_{6,1} + 2x_{6,2} + 2x_{7,2} + 3x_{7,3} + x_{8,1} + 2x_{8,2} + 3x_{8,3} + 2x_{9,2} + 3x_{9,3} + \\ + 4x_{9,4} + x_{10,1} + 2x_{10,2} + 3x_{10,3} + 2x_{11,2} + 3x_{11,3} + 4x_{11,4}$$

Minimum-Latency Scheduling under Resource Constraints (ML-RC)

- Let $\mathbf{t}$ be the vector whose entries are *start times*

$$\mathbf{t} = [t_0, t_1, \dots, t_n]$$

- Formal ILP model

minimize $\mathbf{c}^T \mathbf{t}$ such that

$$\sum_l x_{il} = 1, \quad i = 0, 1, \dots, n$$

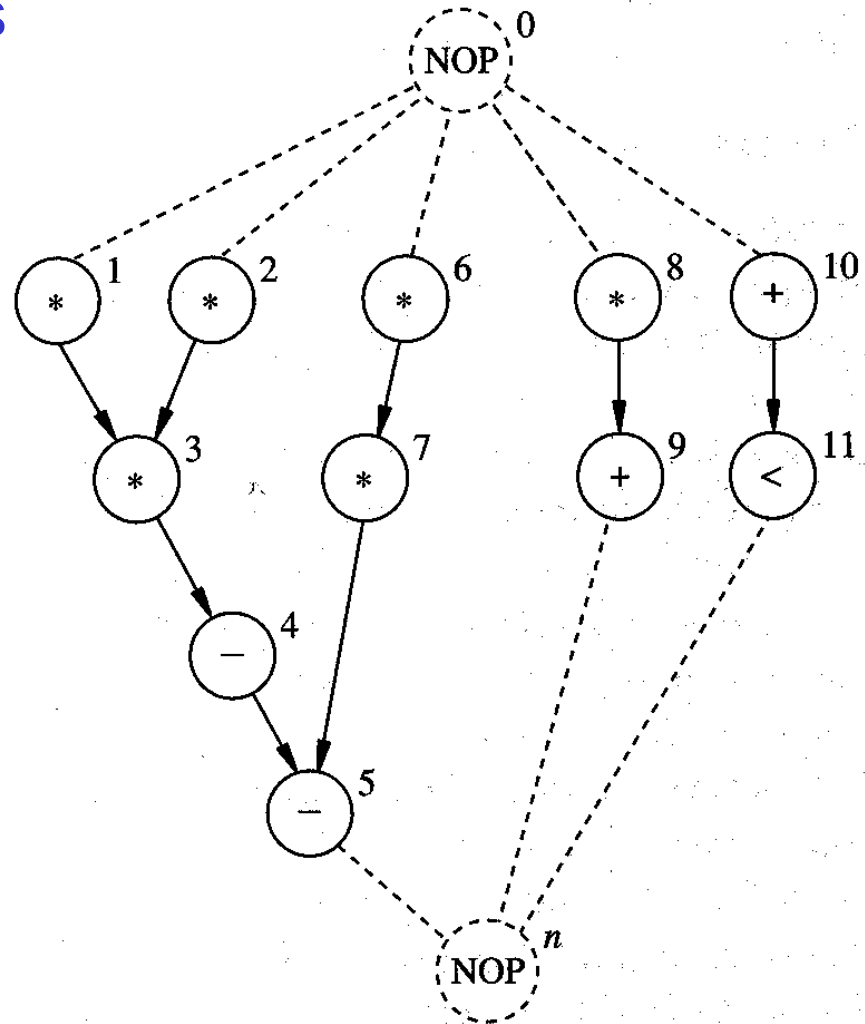
$$\sum_l l \cdot x_{il} - \sum_l l \cdot x_{jl} - d_j \geq 0, \quad i, j = 0, 1, \dots, n : (v_j, v_i) \in E$$

$$\sum_{i: T(v_i)=k} \sum_{m=l-d_i+1}^l x_{im} \leq a_k, \quad k = 1, 2, \dots, n_{res}, \quad l = 1, 2, \dots, \bar{\lambda} + 1$$

$$x_{il} \in \{0, 1\}, \quad i = 0, 1, \dots, n, \quad l = 1, 2, \dots, \bar{\lambda} + 1$$

Example 1 – multiple resources

- Two types of resources
 - MULT
 - Adder, Subtractor
 - Comparator
 - ALU
- Each take 1 cycle of execution time
- Assume upper bound on latency, $L = 4$
- Use ALAP and ASAP to derive bounds on start times for each operator



Example 1 (cont'd.)

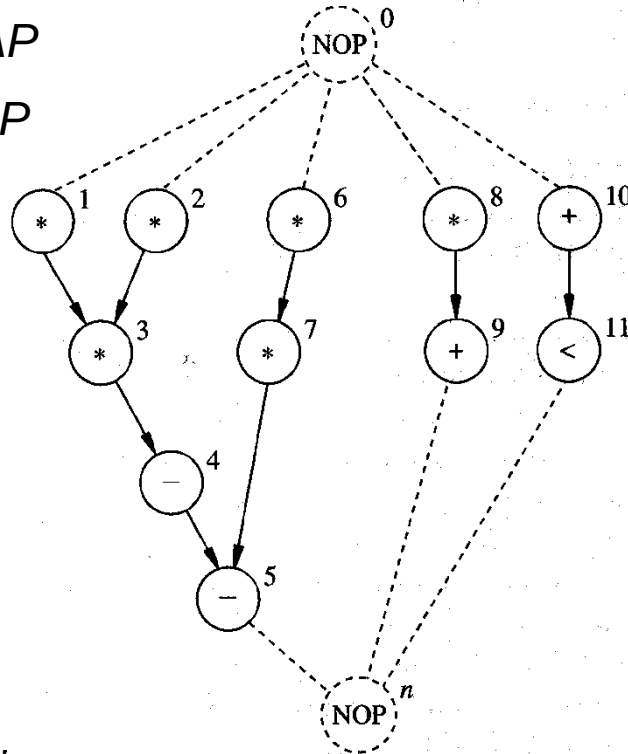
- Start time must be unique

Recall:
$$\sum_l x_{il} = \sum_{l=t_i^S}^{l=t_i^L} x_{il}$$

where:

$t_i^S = t_i$ computed with ASAP

$t_i^L = t_i$ computed with ALAP



$$x_{0,1} = 1$$

$$x_{1,1} = 1$$

$$x_{2,1} = 1$$

$$x_{3,2} = 1$$

$$x_{4,3} = 1$$

$$x_{5,4} = 1$$

$$x_{6,1} + x_{6,2} = 1$$

$$x_{7,2} + x_{7,3} = 1$$

$$x_{8,1} + x_{8,2} + x_{8,3} = 1$$

$$x_{9,2} + x_{9,3} + x_{9,4} = 1$$

$$x_{10,1} + x_{10,2} + x_{10,3} = 1$$

$$x_{11,2} + x_{11,3} + x_{11,4} = 1$$

$$x_{n,5} = 1$$

Example 1 (cont'd.)

- Precedence constraints
 - Note: only non-trivial ones listed

$$2x_{7,2} + 3x_{7,3} - x_{6,1} - 2x_{6,2} - 1 \geq 0$$

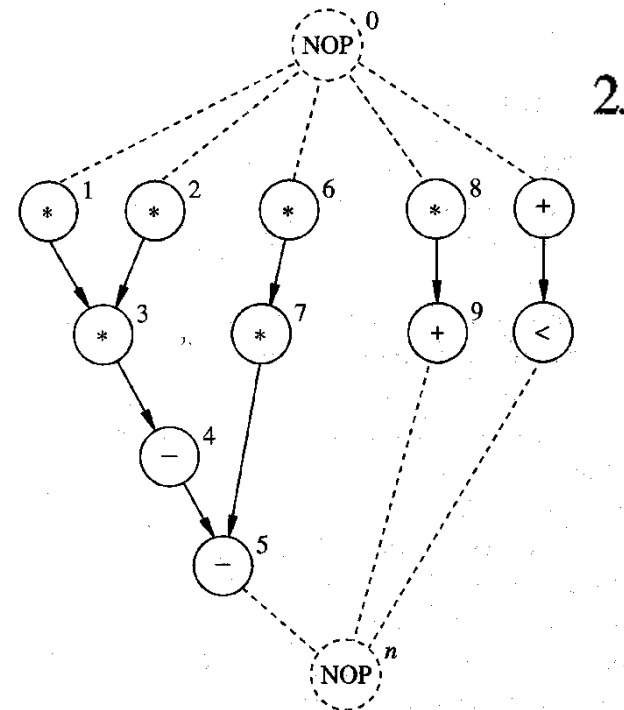
$$2x_{9,2} + 3x_{9,3} + 4x_{9,4} - x_{8,1} - 2x_{8,2} - 3x_{8,3} - 1 \geq 0$$

$$2x_{11,2} + 3x_{11,3} + 4x_{11,4} - x_{10,1} - 2x_{10,2} - 3x_{10,3} - 1 \geq 0$$

$$4x_{5,4} - 2x_{7,2} - 3x_{7,3} - 1 \geq 0$$

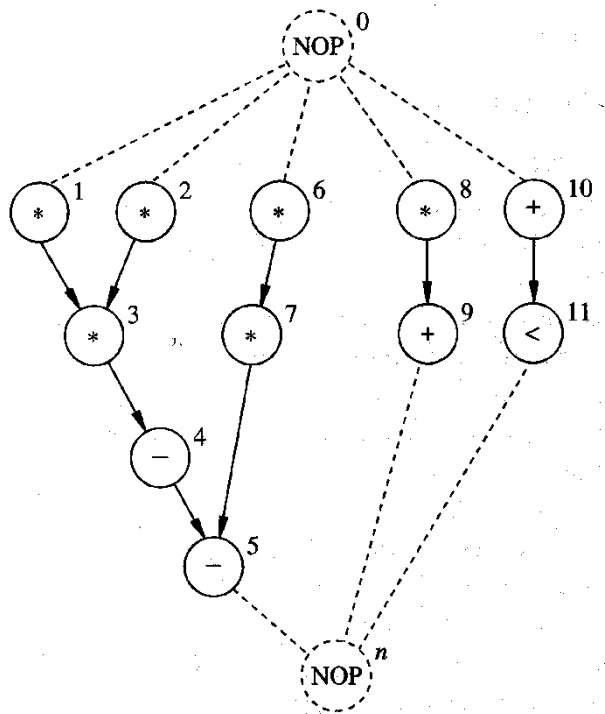
$$5x_{n,5} - 2x_{9,2} - 3x_{9,3} - 4x_{9,4} - 1 \geq 0$$

$$5x_{n,5} - 2x_{11,2} - 3x_{11,3} - 4x_{11,4} - 1 \geq 0$$



Example 1 (cont'd.)

- Resource constraints



MULT
 $a1=2$

ALU
 $a2=2$

$$\left\{ \begin{array}{l} x_{1,1} + x_{2,1} + x_{6,1} + x_{8,1} \leq 2 \\ x_{3,2} + x_{6,2} + x_{7,2} + x_{8,2} \leq 2 \\ x_{7,3} + x_{8,3} \leq 2 \end{array} \right.$$

$$\left\{ \begin{array}{l} x_{10,1} \leq 2 \\ x_{9,2} + x_{10,2} + x_{11,2} \leq 2 \\ x_{4,3} + x_{9,3} + x_{10,3} + x_{11,3} \leq 2 \\ x_{5,4} + x_{9,4} + x_{11,4} \leq 2 \end{array} \right.$$

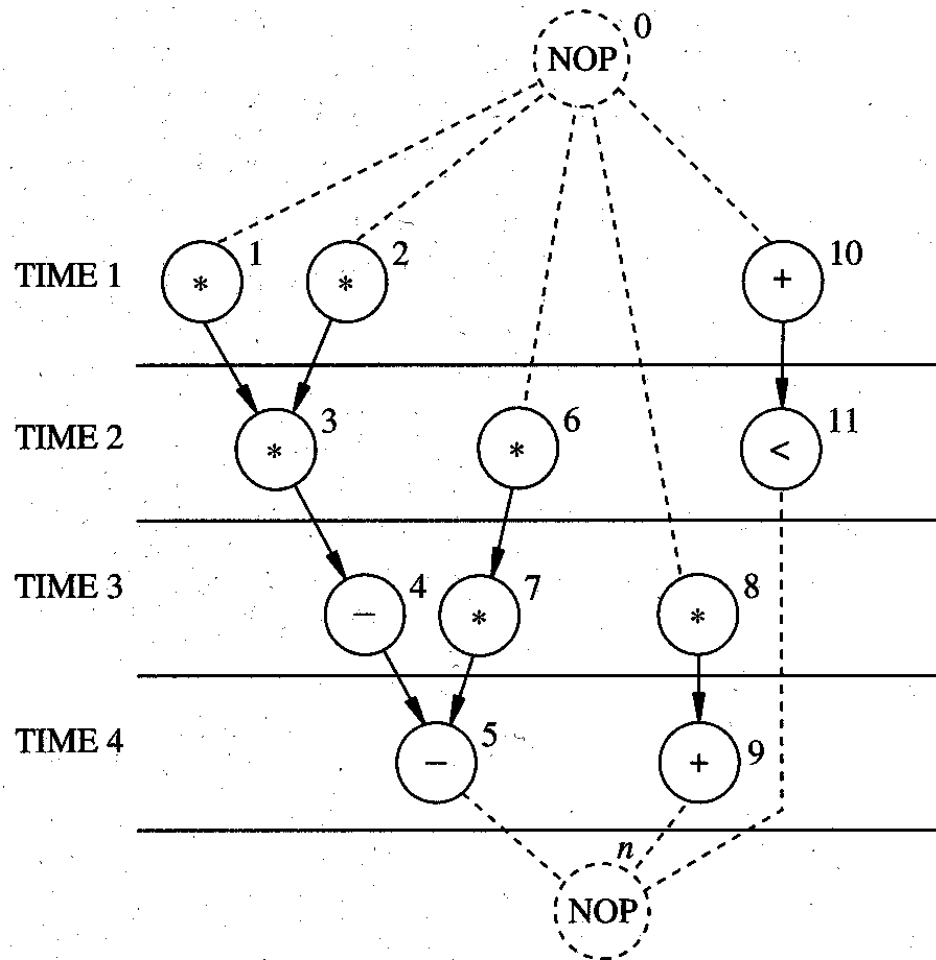
Example 1 (cont'd.)

- Objective function (some possibilities): $F = c^T t$
- $F1: c = [0, 0, \dots, 1]^T$
 - where last entry refers to the *last node* (NOP) !
 - Minimum latency schedule
 - since sink has no mobility ($x_{n,5} = 1$), any feasible schedule is optimum
- $F2: c = [1, 1, \dots, 1]^T$
 - finds earliest start times for all operations $\sum_i \sum_j x_{ij}$
 - or equivalently:

$$x_{6,1} + 2x_{6,2} + 2x_{7,2} + 3x_{7,3} + x_{8,1} + 2x_{8,2} + 3x_{8,3} + 2x_{9,2} + 3x_{9,3} + \\ + 4x_{9,4} + x_{10,1} + 2x_{10,2} + 3x_{10,3} + 2x_{11,2} + 3x_{11,3} + 4x_{11,4}$$

Example Solution 1:

Min. Latency Schedule Under Resource Constraint



Minimum Resource Scheduling under Latency Constraint (MR-LC)

- Special case
 - Identical operations, each executing in one cycle time
- Given a set of operations $\{v_1, v_2, \dots, v_n\}$,
 - find the *minimum number* of operation units needed to complete the execution in k control steps (*MR-LC problem*)
- Integer Linear Programming (ILP):
 - Let y_0 be an integer variable (# units to be minimized)
 - for each control step $l = 1, \dots, k$, define variable x_{il} as

$$x_{il} = \begin{cases} 1, & \text{if computation } v_i \text{ is executed in the } l\text{-th control step} \\ 0, & \text{otherwise} \end{cases}$$
 - define variable y_l (number of units in control step l) for each l

$$y_l = x_{1l} + x_{2l} + \dots + x_{nl} = \sum_i x_{il}$$

ILP Scheduling – simple MR-LC

- Minimize: y_0

Subject to:

- Each computation v_i can start only on $\sum_l x_{il} = 1, i = 0, 1, \dots, n$
 $x_{il} = 1$ for only one value of l (control step) (“vertical” constraint)
- For each precedence relation:
 - If v_j has to be executed after v_i

$$x_{j1} + 2x_{j2} + \dots + kx_{jk} \geq x_{i1} + 2x_{i2} + \dots + kx_{ik} + d(i)$$
- $y_i < y_0$ for all $i = 1, \dots, k$ (steps)

- Meaning of y_0 :
upper bound on the number of units, to be minimized

Example 2 - Formulation

$n = 6$ computations

$k = 3$ control steps

$d(i) = 1$

- Execution constraints:

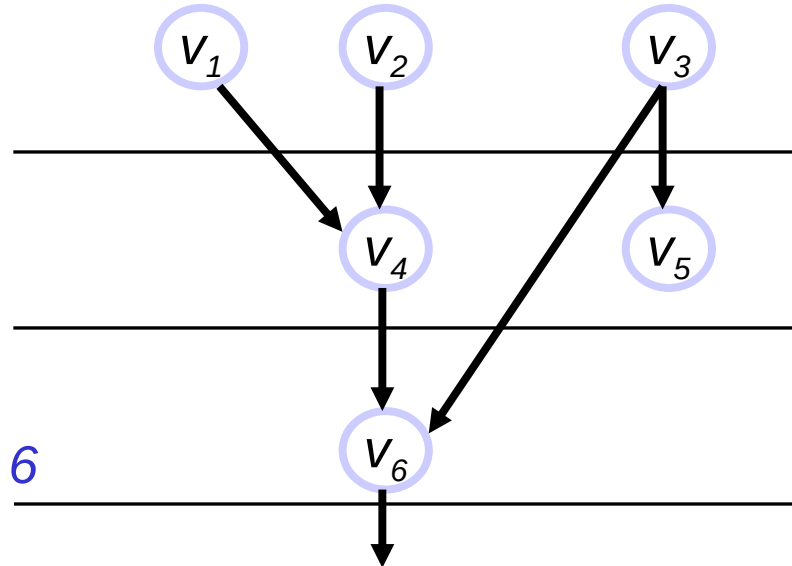
$$x_{i1} + x_{i2} + x_{i3} = 1 \quad \text{for } i = 1, \dots, 6$$

- Resource constraints:

$$y_l = x_{1l} + x_{2l} + x_{3l} + x_{4l} + x_{5l} + x_{6l} \quad \text{for } l = 1, \dots, 3 \text{ (steps)}$$

- Dependency constraints: e.g. v_4 executes after v_1

$$x_{41} + 2x_{42} + 3x_{43} \geq x_{11} + 2x_{12} + 3x_{13} + 1$$



Example 2 – Simplified Formulation

Minimize y_0

- Execution constraints:

$$x_{31} + x_{32} = 1$$

$$x_{52} + x_{53} = 1$$

$$x_{11}, x_{21}, x_{42}, x_{63} = 1$$

- Dependency constraints:

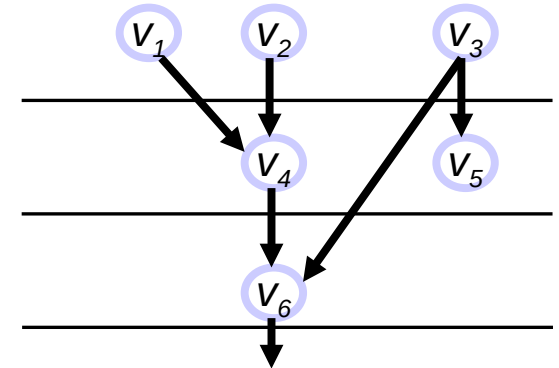
$$2x_{42} \geq x_{11} + 1$$

$$2x_{42} \geq x_{21} + 1$$

$$3x_{63} \geq 2x_{42} + 1$$

$$3x_{63} \geq x_{31} + 2x_{32} + 1$$

$$2x_{52} + 3x_{53} \geq x_{31} + 2x_{32} + 1$$



- Resource constraints:

$$y_1 = x_{11} + x_{21} + x_{31}$$

$$y_2 = x_{42} + x_{32} + x_{52}$$

$$y_3 = x_{63} + x_{53}$$

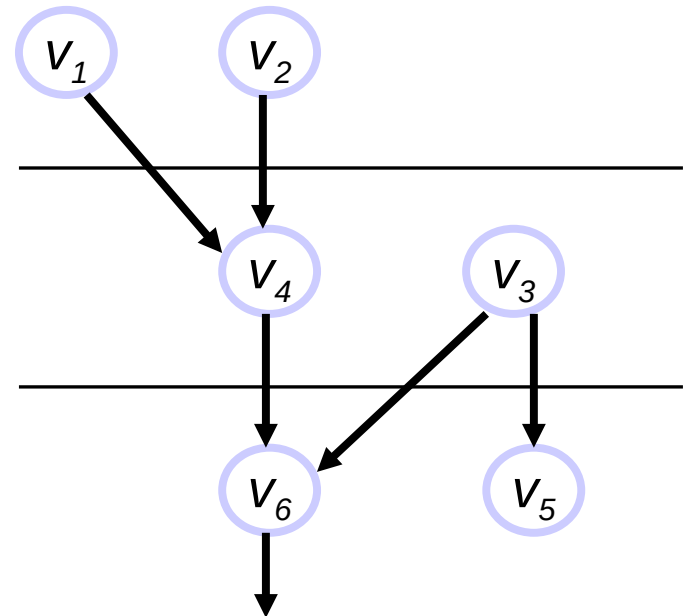
$$y_1 \leq y_0$$

$$y_2 \leq y_0$$

$$y_3 \leq y_0$$

Example 2 - Solution

- Minimize: y_0
- Subject to:
 - $y_l \leq y_0$ for $l = 1, \dots, 3$
 - Starting time constraints ...
 - Precedence constraints ...
- One possible solution:
 - $y_0 = 2$
 - $x_{11} = 1, x_{21} = 1,$
 - $x_{32} = 1, x_{42} = 1,$
 - $x_{53} = 1, x_{63} = 1.$
 - all other $x_{il} = 0$



Minimum Resource Scheduling under Latency Constraint – general MR-LC

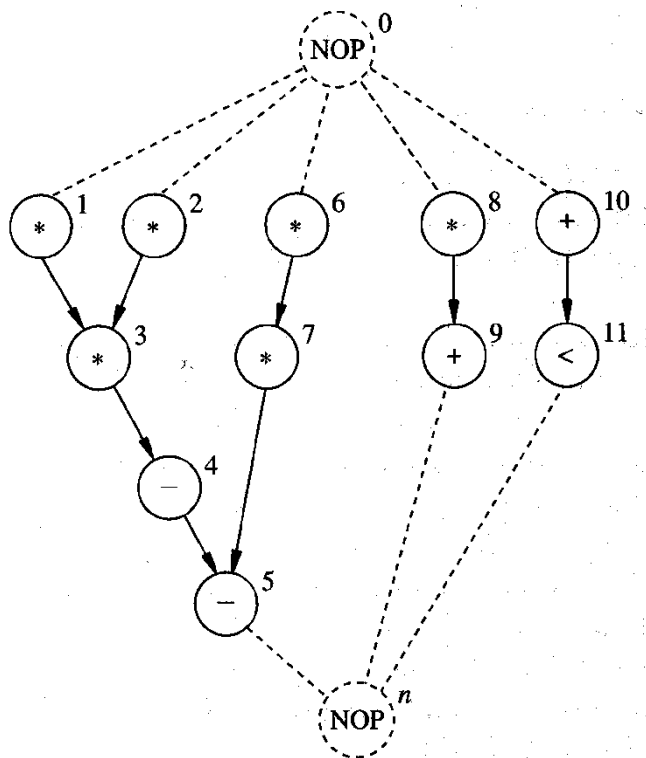
- General case: several operation units (resources)
- Given
 - vector $c = [c_1, \dots, c_r]$ of resource costs (areas)
 - vector $a = [a_1, \dots, a_r]$ of number of resources (unknown)
- Minimize total cost of resources
$$\min \mathbf{c}^T \mathbf{a}$$
- Resource constraints are expressed in terms of variables $a_k = \text{number of operators of type } k$

Example 3 – Min. Resources under Latency Constraint

- Let $c = [5, 1]$
 - MULT costs = 5 units of area, $c_1 = 5$
 - ALU costs = 1 unit of area, $c_2 = 1$
- Starting time constraint – as before
- Sequencing constraints - as before
- Resource constraints – similar to ML-RC, but expressed in terms of *unknown* variables a_1 and a_2
 - a_1 = number of multipliers
 - a_2 = number of ALUs (add/sub)
- Objective function:
 - $$\mathbf{c}^T \mathbf{a} = 5 \cdot a_1 + 1 \cdot a_2$$

Example 3 (contd.)

- Resource constraints



MULT

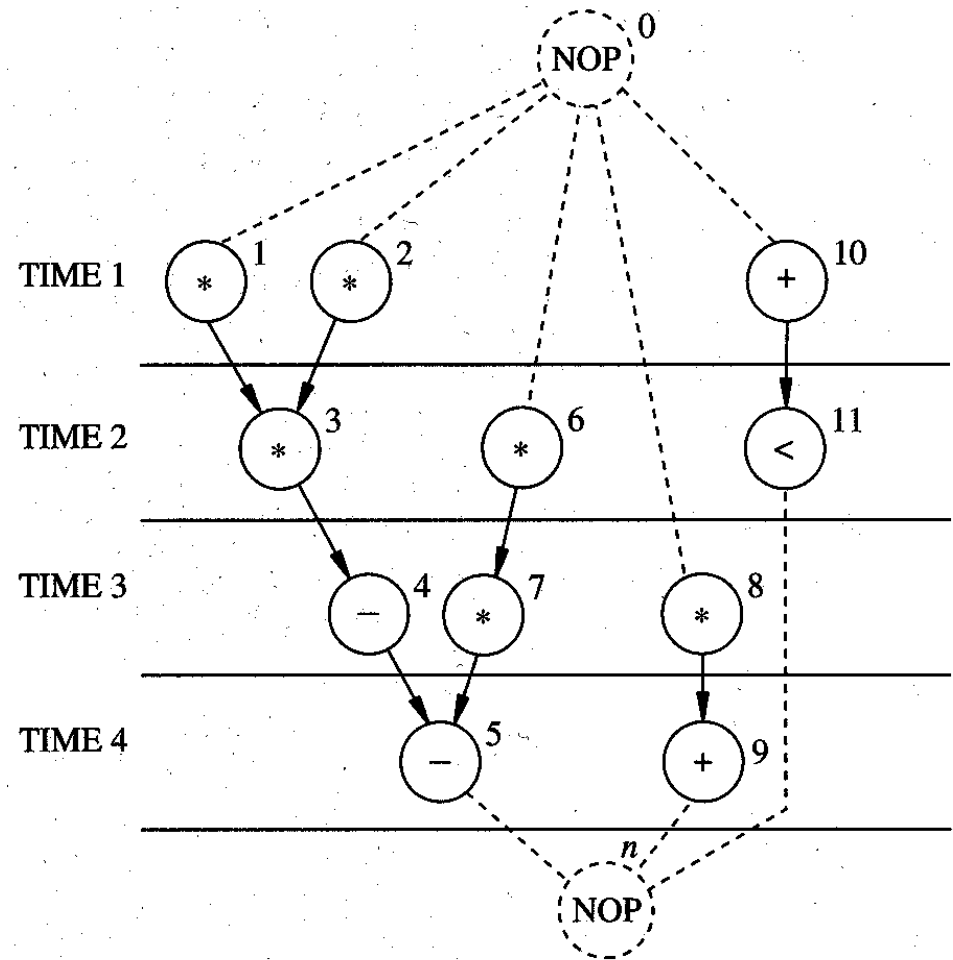
$$\left\{ \begin{array}{l} x_{1,1} + x_{2,1} + x_{6,1} + x_{8,1} - a_1 \leq 0 \\ x_{3,2} + x_{6,2} + x_{7,2} + x_{8,2} - a_1 \leq 0 \\ x_{7,3} + x_{8,3} - a_1 \leq 0 \end{array} \right.$$

ALU

$$\left\{ \begin{array}{l} x_{10,1} - a_2 \leq 0 \\ x_{9,2} + x_{10,2} + x_{11,2} - a_2 \leq 0 \\ x_{4,3} + x_{9,3} + x_{10,3} + x_{11,3} - a_2 \leq 0 \\ x_{5,4} + x_{9,4} + x_{11,4} - a_2 \leq 0 \end{array} \right.$$

Example 3 - Solution

- Minimize
 $\mathbf{c}^T \mathbf{a} = 5 \cdot a_1 + 1 \cdot a_2$
- Solution with $\text{cost} = 12$
 $a_1 = 2$
 $a_2 = 2$



Precedence-constrained Multiprocessor Scheduling

- All operations performed by the same type of resource
 - intractable problem; even if operations have unit delay
 - except when the G_c is a tree (then it is optimal and $O(n)$)
 - *Hu's algorithm*

minimize $\mathbf{c}^T \mathbf{t}$ such that

$$\sum_l x_{il} = 1, \quad i = 0, 1, \dots, n$$

$$\sum_l l \cdot x_{il} - \sum_l l \cdot x_{jl} \geq 1, \quad i, j = 0, 1, \dots, n : (v_j, v_i) \in E$$

$$\sum_i x_{il} \leq a, \quad l = 1, 2, \dots, \bar{\lambda} + 1$$

$$x_{il} \in \{0, 1\}, \quad i = 0, 1, \dots, n, \quad l = 1, 2, \dots, \bar{\lambda} + 1$$